Migrating from ST25R39xx to ST25R500 and ST25R300

Introduction

This application note provides the reader with a list of considerations and requirements to ensure a successful migration from ST25R3911B, ST25R3916, or ST25R3916B (referred to as ST25R39xx) to ST25R300 and ST25R500 multipurpose NFC transceivers.

The device delivers high-end performance in a 5x5 mm package. The NFC transceiver brings the convenience of contactless interaction and features to a variety of end applications. It is especially optimized for end products in the commercial and automotive sectors.

ST25R automotive series NFC readers enable smart and convenient car access and start functionality, being compliant with the CCC & NFC Forum standards.

Specifically designed for consumer and industrial end products, the transceiver is a versatile solution for enabling NFC use cases. With its compact size and advanced features, it is an ideal choice for automotive manufacturers looking to incorporate NFC technology into their products.

This document applies to the ST25R500 and its derivative, the ST25R300.

For further information on ST25R3911, ST25R3916, and ST25R500, refer to the documents in Table 2.

Table 1. Applicable products

Type	RPNs
NFC transceivers	ST25R3911B, ST25R3916, ST25R3916B ST25R300, ST25R500

1 General information

Table 2. **Referenced documents**

N°	Document reference	Document title
[1]	DS11793	ST25R3911B datasheet
[2]	DS13541	ST25R3916B(1) datasheet
[3]	DS14655	ST25R300 datasheet
[4]	DS14593	ST25R500 datasheet
[5]	AN6279	Layout recommendations for the design of boards with the ST25R300 and ST25R500 device
[6]	AN6298	Wake-up mode for the ST25R300 and ST25R500
[7]	AN6092	Antenna design for ST25R500 and ST25R300 devices

1. *In the scope of this document, the ST25R3916 and ST25R3916B are considered identical. The ST25R3916B features advanced AWS compared to the ST25R3916, which is not relevant for the migration of either device.*

Table 3. **Acronyms**

Acronym	Definition
AP2P	Active peer to peer
IRQ	Interrupt request
MCU	Microcontroller unit
P2P	Peer to peer
ACM	Active communication mode (AP2P)
PCM	Passive communication mode
PCB	Printed circuit board
PSRR	Power supply rejection ratio
RF	Radio frequency
RFAL	Radio frequency abstraction layer
HAL	Hardware abstraction layer
WU	Wake-up
MRT	Mask receive timer
NRT	No response timer
GPT	General-purpose timer
WUT	Wake-up timer
WPT	Wireless power transfer

2 Pinout comparison

ST25R39xx, ST25R300, and ST25R500 devices are all available in QFN32 packages.

All devices are 32 pin packages (5 x 5 mm). Additional features introduced in the devices require a pinout change, which breaks the chips pin-to-pin compatibility.

Note: For further information, refer to the application note [5].

Table 4. Pinout comparison between ST25R3911B and ST25R500

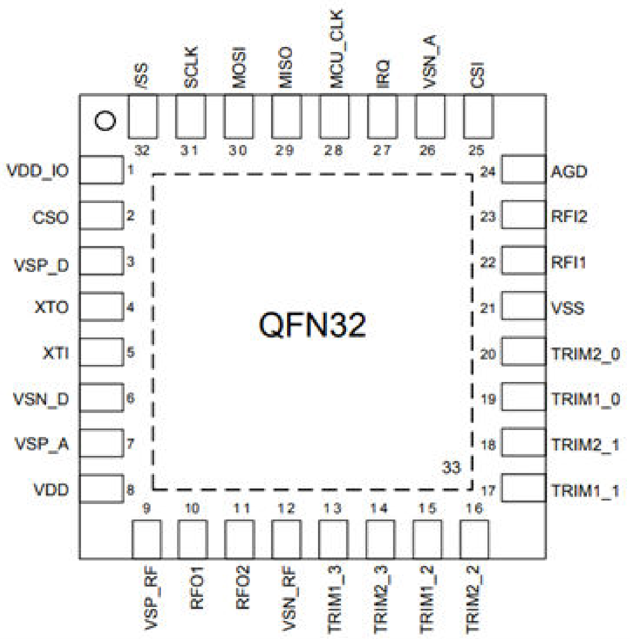
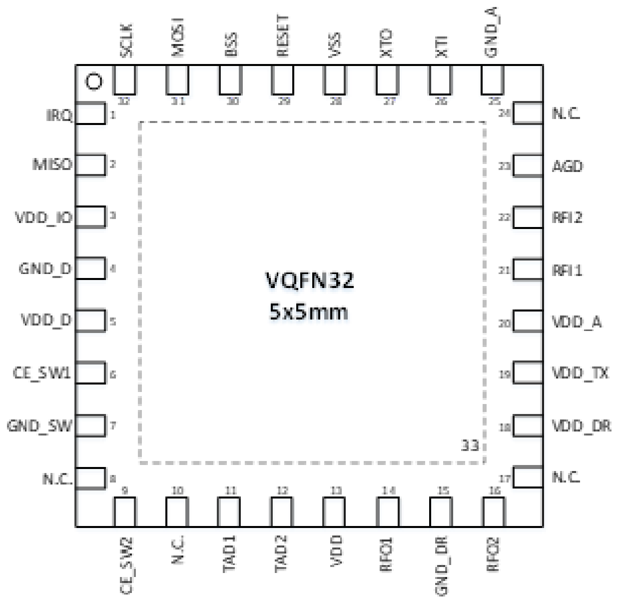
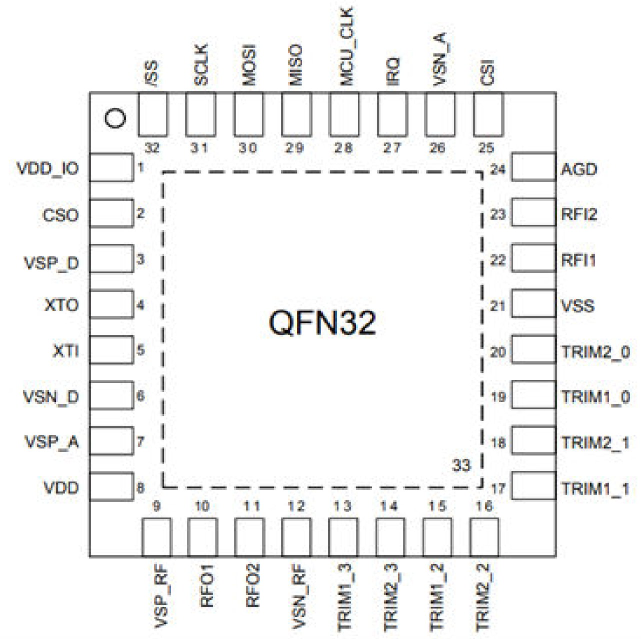
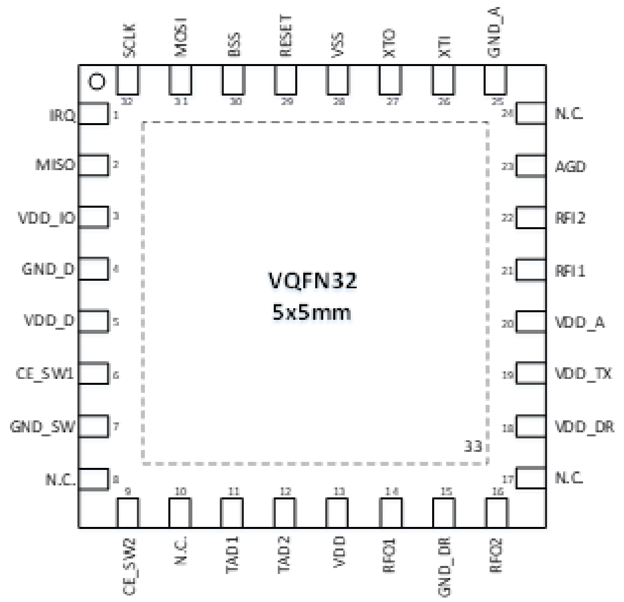
<table>
<tr><th>ST25R3911B</th><th>ST25R500</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

Table 5. **Pinout comparison between ST25R3916(B) and ST25R500**

<table>
<tr><th>ST25R3916(B)</th><th>ST25R500</th></tr>
<tr><td>


</td><td>


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</table>

The Table 6 describes the main pinout differences.

Table 6. **Pinout differences**

Element	Description
Reset pin	The ST25R500 introduces a reset pin (active high) with the possibility to perform a reset via an external pin. This pin can be used to put the device in the lowest power consumption mode, when the NFC functionality is not needed. This pin is optional and must be tied to GND when it is not used. The device can also be reset via direct command, as the previous devices.
Capacitive sensing pins	On ST25R500, capacitive sensing is no longer available, and only inductive measurements are available. Due to that reason, the CSI/CSO pins are no longer present.
Exposed pad	On ST25R500, pin 33 (exposed pad) is not only used for thermal release, but also to connect to GND. A good connection to the PCB ground plane via multiple vias is extremely important to ensure proper operation.
Clock output for MCU	Not available on ST25R500.
I2C interface enable	ST25R500.supports Serial peripheral interface and thus I2C_EN is not available on ST25R500.
Tune voltage for AAT	Not available on ST25R500.
Input to trim antenna resonant circuit	Not available on ST25R500.
External load modulation	The ST25R500 supports different passive load modulation methods through GPIO1 or GPIO1/GPIO2, or GPIO1/GPIO2/TAD1/TAD2. Due to this reason the EXT_LM is no longer present. See *Passive load modulation* section in the ST25R500 datasheet for detailed information.
Supplies	The ST25R500 is designed to operate from a wide power supply range and a wide peripheral I/O voltage range. See *Operating conditions* section in the datasheet of your device for detailed information.
Ground	ST25R500 pins mentioned as N.C. in the datasheet are not bonded, they must be connected to the PCB ground.

3 Features description and comparison

This section lists and compares the main features of the devices.

3.1 Power management

The ST25R500 features three positive supply pins: VDD, VDD_TX, and VDD_IO. VDD is the main power supply pin, supplying the analog and digital blocks through two regulators (VDD_A, VDD_D). VDD_TX is the transmitter power supply pin. See the *Operating Conditions* section in the datasheet of your device for detailed values of the voltage range. VDD and VDD_TX should be directly connected, the voltage difference between VDD and VDD_TX should not be greater than 0.2 V.

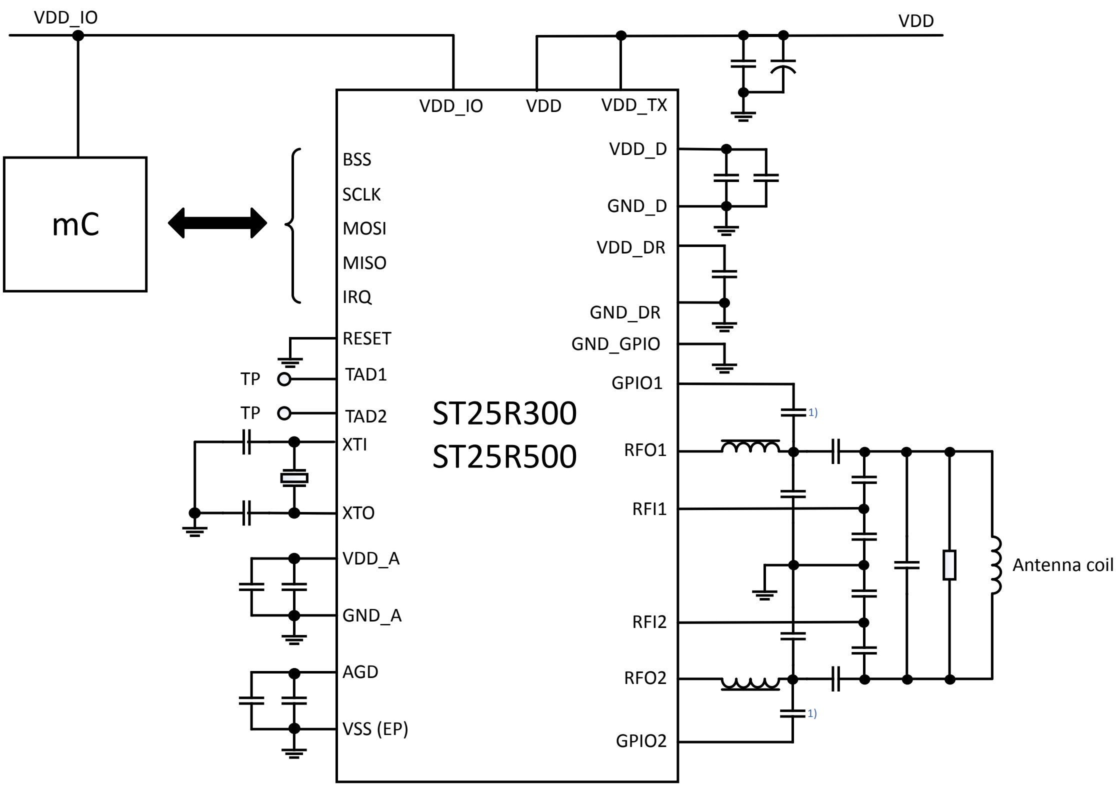
The regulated voltage can be automatically adjusted to achieve the maximum possible regulated voltage while maintaining good PSRR. All regulator pins also have corresponding negative supply pins (except VDD_TX), which are externally connected to the ground potential. The recommended blocking capacitors for supply pins are as follows:

- VDD, VDD_TX, VDD_D, VDD_A (pins 13, 19, 5, 20): 2.2µF in parallel with 10 nF.
- VDD_IO (pin 3): 100 nF.
- VDD_DR (pin 18): 47 nF.
- AGD (pin 23): 1µF in parallel with 10 nF.

On the ST25R500, the regulation of VDD_DR can be set by a defined voltage configuration or by a target voltage drop in reference to the voltage supplied at VDD. A power-down support block is introduced to maintain low-power consumption and support VDD_D in PD mode, where the regulators are disabled.

For further information, refer to the datasheet [4].

Figure 1. System diagram – differential antenna driving



1) Only required for CE operation. If only RW operation is required, the capacitor can be D.N.P and the pin left open.

3.2 Automatic antenna tuning (AAT)

Similar to the ST25R3911B and ST25R3916, the ST25R300 and ST25R500 can change the characteristics of the antenna matching during operation. On these devices, antenna tuning is supported via external MOS transistors and capacitors, driven by GPIO1/GPIO2 and TAD1/TAD2 or by switching the EMC filter cutoff frequency via GPIO pins. Further details about both methods can be found in the datasheet of the ST25R300 or ST25R500.

3.3 Serial interface

The ST25R300 and the ST25R500 use SPI as an interface to hosts, with a speed of up to 10 Mbit/s. Any system or device communicating with the ST25R3911B or ST25R3916 (with respective speeds of up to 6 Mbit/s and 10 Mbit/s) via SPI must be able to interface with the new devices if speed matching is done correctly and data is correctly sampled on the falling edge of SCLK. The maximum SPI speed of the ST25R300 and ST25R500 is 5 Mbit/s if the VDD_IO supply voltage is in a range of 1.65 V to 2.7 V and 10 Mbit/s if the VDD_IO supply voltage is in a range of 2.7 V to 5.5 V.

The MISO output is usually in a tristate on the ST25R300 and the ST25R500, and is only driven when output data is available. Due to this, the MOSI and the MISO can be externally shorted to create a bidirectional signal. During the time the MISO output is in a tristate, it is possible to switch on a 10 kΩ pull-down by activating option bits \`miso_pd1\` and \`miso_pd2\`.

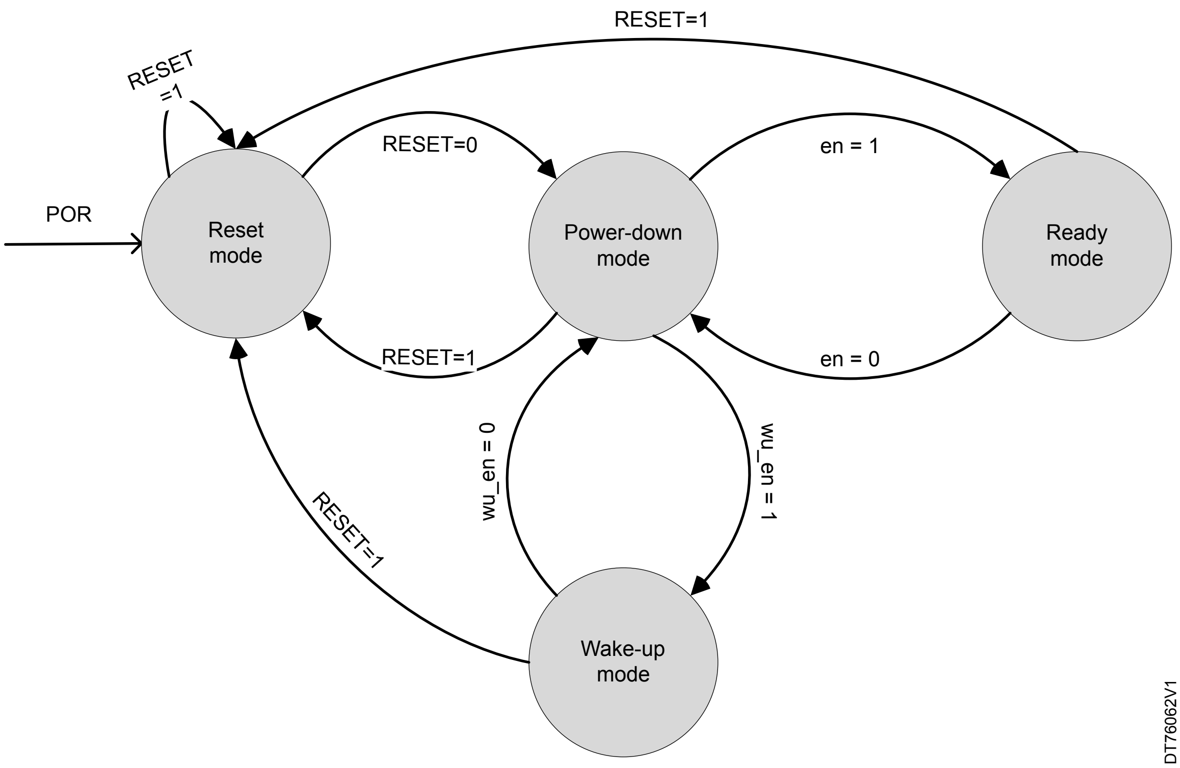
Moreover, the devices do not feature I2C as an interface to hosts, unlike the ST25R3916, which features both SPI and I2C.

3.4 Operating modes

The device introduces a new mode controlled by the RESET pin. It features four operating modes:

- **Reset Mode (RESET):** When the device is powered and the RESET pin is high, the device enters reset mode. All blocks are deactivated, no clock is present, and power consumption is minimized.
- **Power-down Mode (PD):** At power-up, when the RESET pin is low, the device enters power-down mode. In this mode, the AFE's static power consumption is very low, but SPI communication and register access in the PD domain are possible.
- **Wake-up Mode (WU):** In this mode, the device checks for changes in antenna properties at regular intervals and triggers an IRQ if a change is detected.
- **Ready Mode (RD):** In this mode, the regulators and crystal oscillator are enabled, and the device is ready to transmit and receive.

Figure 2. **Transition from one state to another**



3.5 Consumption comparison

The following table compares the average consumption of the available modes for the ST25R3911B, ST25R3916, and ST25R500 and its derivative.

Table 7. **Typical power consumption comparison (VDD = 3.3 V, 25°C)**

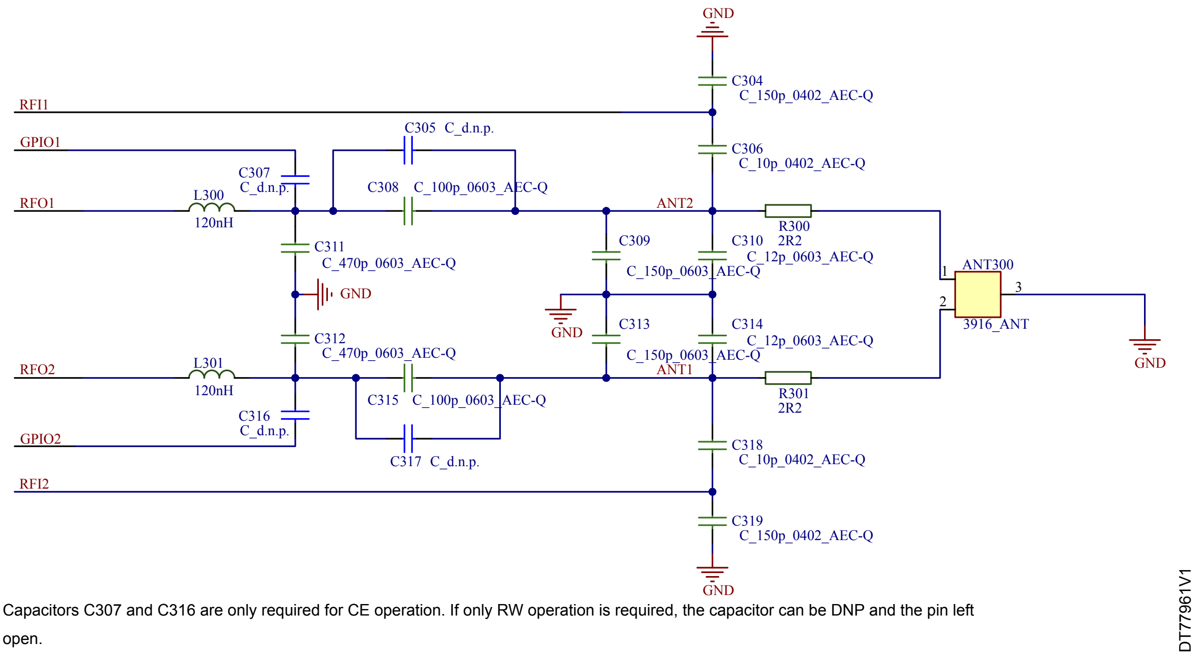
Operating condition	ST25R3911B	ST25R3916	ST25R500
Reset	-	-	0.038 µA
Power-down	0.7 µA	0.8 µA	1.4 µA
Wake-up	3.6 µA	3.0 µA	2.0 µA
Card emulation	-	3.5 µA	4.6 µA
Ready	5.4 mA	4.5 mA	2.8 mA
all active	8.7 mA	16 mA	11.7 mA

Note: See Electrical specification section in the datasheet of your device for the configuration of each operating condition. The modes from the ST25R500 and its derivative can have similarities in their functionalities or the way they work with the modes of the ST25R39xx, but their configurations are different. For further information, refer to the ST25R500 datasheet [4].

3.6 Matching circuit

The ST25R500 introduces GPIO1 and GPIO2, which can be used when card emulation is utilized, and NFC Forum analog Listener or ISO10373-6 compliance is required. The minimum configuration of the matching circuit for the ST25R500 is depicted in Figure 3.

Figure 3. ST25R500 matching circuit minimum configuration



Capacitors C307 and C316 are only required for CE operation. If only RW operation is required, the capacitor can be DNP and the pin left open.

The steps to design the matching circuit for the ST25R500 are explained in the corresponding application note [7]. It is strongly recommended to use the antenna matching tool (STSW-ST25R004), which helps calculate the tuning components and develop the initial matching circuit for the ST25R500.

3.7 Wake-up mode

The wake-up (WU) mode on the ST25R500 and its derivatives is distinct from that of the ST25R39xx. Table 8 highlights the differences in their wake-up mode features:

Table 8. Wake-up mode comparison

Wake-up mode features	ST25R39xx	ST25R500
Capacitive sensing	Yes	No
Inductive sensing	AM/PM channels	I/Q decomposition
Calibration required	No	Yes
Configurable measurement of pulse duration	No	Yes
Configurable wake-up trigger criteria	No	Yes

3.8 Supported NFC technologies and modes

The device supports most of the existing technologies and modes anticipated by the NFC Forum, except for:

- Card emulator in NFC-B
- NFC-ACM

4 Operation and interface

The device has comparable operation modes and interfaces to previous ST25R39xx multipurpose NFC transceivers. Despite these similarities, each device has its own specific features, with some added or removed. It is recommended to use the STMicroelectronics NFC library (RFAL), which supports all the required technologies and protocols while encapsulating all existing ST25Rxxxx devices (such as ST25R3911B, ST25R3916, ST25R95, ST25R200, and ST25R500). Tweaking outdated ST25R drivers is not recommended. If RFAL is already in use for the project, migrating from ST25R3911B or ST25R3916 to ST25R500 and its derivatives is as simple as switching the drivers to compile. Further details are provided in the following sections.

4.1 Interface scheme

Compared to the ST25R39xx, the device uses a different addressing scheme. The ST25R500 scheme allows for:

- 88 addressable registers (0x00 - 0x57)
- 31 command codes (0x60-0x7F and 0xE0-0xFF except 0xFC/0xFD)
- FIFO and CE memory access (0x5F, 0x58, and 0x5A)
- Test/OTP registers access via a dedicated command

For further information, refer to the *SPI* section of the datasheet [4].

4.2 Direct commands

The ST25R500 features 24 direct commands. Although some commands may share the same code, several new commands and features have been added, while others have been removed. The driver file `st25r500.h` provides various definitions for the direct commands. For further information, refer to the *Direct Commands* section of the datasheet [4].

4.3 Registers

Two types of registers are available in the device:

- Configuration registers: These can be read and written (RW) through the SPI and hold the device's configuration.
- Display registers: These are read-only (RO) and contain information about the device’s internal state.

Additionally, registers are grouped into two different power domains:

- Power down (PD)
- Ready (RD)

Several new registers are added to the device to accommodate new features, and some configuration bits have been moved. No exhaustive list of the changes is provided, as very few registers maintain the same address and content between devices. The driver file `st25r500_com.h` provides various definitions for the registers.

For further information, refer to the *Register* section of the ST25R500 datasheet [4].

Note: *Even though several configuration and bitfields are similar and have analogous functionality, the addresses of the registers on the ST25R500 and it derivatives have no direct mapping to the ST25R39xx device.*

4.4 FIFO

Reading received data from the FIFO is similar to reading data from the addressable registers. Read and write operations are performed by accessing address 0x5F, resulting in 0x5F for a write operation and 0xDF for a read operation.

The device has a FIFO of 256 bytes, but the transmission and reception capabilities of the device are not limited by the FIFO size, thanks to a FIFO water level of 128 bytes in both directions (transmission and reception).

4.5 Interrupts

The device maintains an almost identical interrupt interface with an IRQ pin and three IRQ registers. However, the passive target interrupt register found in the ST25R3916 is absent from the ST25R500 and its derivatives. Some IRQs retain the same functions as on the ST25R39xx but are located at different addresses.

Interrupt signals on this device are mostly identical to those on the ST25R39xx, with the following exceptions:

Interrupt Signal	Description
I_subc_start	New IRQ due to subcarrier start.
I_hfe	Renamed from I_err1, indicates a hard framing error.
I_sfe	Renamed from I_err2, indicates a soft framing error.
I_wuq	New IRQ indicating wake-up Q-channel measurement.
I_wui	New IRQ indicating wake-up I-channel measurement.
I_wut	Renamed from I_wt, indicates a Wake-up timer expiry.
I_wutme	New IRQ after each WU measurement event.
I_wpt_stop	New IRQ due to WPT Stop detection.
I_wpt_fod	New IRQ due to WPT Foreign object detection.
I_wam/I_wph	Removed. See I_wui/I_wuq.
I_ce_sc	New IRQ due to CE state change.
I_rxe_cea	Renamed from I_rxe_pta.

For further details, refer to the ST25R500 datasheet [4].

4.6 Timers

The timer structure on the ST25R39xx and ST25R300 and ST25R500 is similar and includes the following timers:

- MRT (Mask Receive Timer)
- NRT (No Response Timer)
- GPT (General Purpose Timer)
- WUT (Wake-up Timer)

However, the MRT in the ST25R500 allows additional steps for increased precision.

5 Migrating existing software projects

The use of the STMicroelectronics NFC library (RF abstraction layer) is strongly recommended for this device. RFAL supports all required technologies and protocols while encapsulating all existing ST25Rxxxx devices, including ST25R3911B, ST25R3916, ST25R95, ST25R200, and ST25R500. The STMicroelectronics NFC library is periodically updated to introduce new features, expand NFC device support, and provide fixes. Its compliance with the latest versions of relevant standards is continuously ensured. Patching and tweaking existing non-RFAL ST25Rxxxx drivers is not recommended, as an incomplete driver that cannot be supported is unable to receive official updates.

5.1 RFAL projects

5.1.1 General

The RF abstraction layer (RFAL) is designed to encapsulate the ST25Rxxxx device underneath. Migrating an existing ST25R500 (or any previous ST25Rxxxx) project involves modifying the RFAL HAL to support the ST25R500. Here are the steps:

1. In the project compilation list or makefile, replace the preprocessor instruction defining the targeted ST25R3911B, ST25R3916, or ST25R3916B device with ST25R500.
2. Replace the included folder from `rfal/source/st25r3911` or `st25r3916` to `rfal/source/st25r500`.
3. Switch the relevant modules from ST25R3911B or ST25R3916 to the equivalent for ST25R500:

ST25R3911B	ST25R3916	ST25R500
rfal_rfst25r3911	rfal_rfst25r3916	rfal_rfst25r500
st25r3911	st25r3916	st25r500
st25r3911_com	st25r3916_com	st25r500_com
st25r3911_interrupt	st25r3916_irq	st25r500_irq

To ensure application portability, avoid direct calls to the ST25Rxxxx HAL driver part of the RFAL and refrain from executing device direct commands. Extra porting efforts are needed for existing applications that do not follow these recommendations.

5.1.2 Specific cases

5.1.2.1 NFC-ACM

For applications based on the `rfal_nfc` module, the `RFAL_NFC_POLL_TECH_AP2P` and `RFAL_NFC_LISTEN_TECH_AP2P` technologies must not be used as AP2P (ACM) is not supported.

For applications using RFAL low layer modules, an `RFAL_ERR_NOTSUPP` code is returned when `RFAL_MODE_POLL_ACTIVE_P2P` or `RFAL_MODE_LISTEN_ACTIVE_P2P` is used in `rfalSetMode()`.

5.1.2.2 Low-power mode

Existing applications using the low-power mode can benefit from reduced power consumption by utilizing the `RFAL_LP_MODE_HR` low-power mode, which is available when the RESET pin is controlled by a GPIO.

Note: *When using the RESET pin, the device is reinitialized.*

5.1.2.3 Analog configuration

The device RFAL HAL includes a default analog configuration table in `rfal_analogConfigTbl.h`. Analog configurations from the ST25R39xx should not be reused. If a custom analog configuration was used in an existing ST25R39xx application, it is recommended to derive the new custom analog configuration from the ST25R500 default analog configuration table. The driver file `st25r500_com.h` provides various definitions for the registers and their fields.

5.1.3 Migration from X-CUBE-NFC6

An existing STM32CubeMX generated project based on X-CUBE-NFC6 can be regenerated by replacing X-CUBE-NFC6 with X-CUBE-NFC12.

5.2 Non-RFAL projects

The standard procedure is to use the RFAL, as mentioned in Section 5.1. To adapt your project accordingly, refer to the existing documentation and reference designs available on www.st.com.

Revision history

Table 9. **Document revision history**

Date	Version	Changes
23-Jun-2025	1	Initial release.

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